

## CHARLEVILLE-MEZIERES AP, France

WMO: 070750

Lat: 49.7828N

Lon: 4.6431E

Elev: 149

StdP: 99.55

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -8.2       | -5.9 | -11.3                         | 1.4 | -5.5 | -9.0 | 1.8 | -4.6 | 9.6                   | 7.8  | 8.4 | 7.0  | 0.8                   | 300  | 0.524 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.7                   | 29.7            | 20.5 | 27.7 | 19.7 | 26.0 | 18.8 | 21.5                | 28.0 | 20.4 | 26.4 | 19.4 | 24.7 | 2.8                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.2                            | 14.2 | 24.7 | 18.2 | 13.4 | 23.2 | 17.4 | 12.7 | 22.0 | 63.2          | 28.1 | 59.4 | 26.4 | 56.0 | 24.9 | 26.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
| (a)               | (b)  | (c) | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| 7.5               | 6.6  | 5.9 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
|                   |      |     | -11.6                      | 33.1 | 2.8                | 2.4 | -13.7                                              | 34.8 | -15.3      | 36.2 | -16.9      | 37.6 | -18.9      | 39.3 |
|                   |      |     | -11.9                      | 23.8 | 2.7                | 1.3 | -13.8                                              | 24.7 | -15.4      | 25.4 | -16.9      | 26.1 | -18.9      | 27.0 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.8    | 2.4   | 3.1   | 5.9   | 8.9   | 12.7  | 15.9  | 17.9  | 17.3  | 13.8  | 10.3  | 6.1   | 3.2  |
|                                                                                  | DBStd       | 6.46   | 4.24  | 3.78  | 3.32  | 3.41  | 3.16  | 3.22  | 3.12  | 2.87  | 2.81  | 3.55  | 3.73  | 4.10 |
|                                                                                  | HDD10.0     | 1012   | 236   | 193   | 131   | 61    | 10    | 1     | 0     | 0     | 3     | 39    | 125   | 213  |
|                                                                                  | HDD18.3     | 3176   | 494   | 427   | 385   | 284   | 177   | 86    | 48    | 55    | 137   | 248   | 367   | 469  |
|                                                                                  | CDD10.0     | 950    | 1     | 0     | 4     | 27    | 93    | 179   | 244   | 225   | 118   | 49    | 8     | 2    |
|                                                                                  | CDD18.3     | 73     | 0     | 0     | 0     | 0     | 2     | 14    | 34    | 22    | 2     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 1103   | 0     | 0     | 0     | 13    | 58    | 232   | 430   | 316   | 52    | 2     | 0     | 0    |
|                                                                                  | CDH26.7     | 296    | 0     | 0     | 0     | 1     | 7     | 58    | 128   | 94    | 7     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.5    | 2.9   | 2.9   | 2.9   | 2.6   | 2.4   | 2.3   | 2.2   | 1.9   | 2.0   | 2.3   | 2.6   | 2.8  |
| Precipitation                                                                    | PrecAvg     | 1050   | 116   | 90    | 77    | 67    | 70    | 82    | 83    | 84    | 74    | 93    | 92    | 122  |
|                                                                                  | PrecMax     | 1279   | 281   | 208   | 181   | 182   | 149   | 138   | 185   | 173   | 156   | 213   | 163   | 296  |
|                                                                                  | PrecMin     | 780    | 19    | 17    | 5     | 9     | 9     | 25    | 41    | 25    | 24    | 41    | 19    | 28   |
|                                                                                  | PrecStd     | 140    | 55    | 48    | 42    | 36    | 37    | 37    | 32    | 47    | 36    | 38    | 41    | 59   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 12.6  | 13.9  | 19.5  | 25.2  | 27.6  | 31.2  | 33.0  | 32.9  | 27.6  | 22.8  | 16.3  | 13.2 |
|                                                                                  |             | MCWB   | 11.1  | 9.1   | 11.8  | 15.9  | 19.3  | 21.4  | 21.3  | 21.8  | 19.3  | 17.5  | 13.9  | 11.4 |
|                                                                                  | 2%          | DB     | 11.2  | 11.5  | 16.8  | 22.0  | 25.0  | 28.2  | 29.7  | 29.2  | 24.8  | 19.7  | 14.3  | 11.8 |
|                                                                                  |             | MCWB   | 9.8   | 8.9   | 10.9  | 14.3  | 16.9  | 19.8  | 20.5  | 20.7  | 18.5  | 16.0  | 12.4  | 10.6 |
|                                                                                  | 5%          | DB     | 9.7   | 10.1  | 14.2  | 19.0  | 22.7  | 26.0  | 27.7  | 26.9  | 22.3  | 17.6  | 12.9  | 10.6 |
|                                                                                  |             | MCWB   | 8.5   | 8.5   | 9.7   | 12.9  | 16.1  | 18.8  | 19.6  | 19.6  | 17.1  | 14.8  | 11.5  | 9.5  |
|                                                                                  | 10%         | DB     | 8.4   | 9.0   | 12.1  | 16.7  | 20.4  | 23.8  | 25.7  | 24.6  | 20.0  | 16.0  | 11.6  | 9.2  |
|                                                                                  |             | MCWB   | 7.2   | 7.5   | 9.2   | 11.5  | 14.9  | 17.6  | 18.9  | 18.4  | 15.5  | 13.9  | 10.4  | 8.3  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 11.5  | 10.6  | 12.8  | 16.7  | 20.3  | 22.4  | 22.7  | 23.0  | 20.2  | 18.0  | 14.3  | 11.7 |
|                                                                                  |             | MCDB   | 12.3  | 12.4  | 17.2  | 23.5  | 26.3  | 29.0  | 30.3  | 30.6  | 25.8  | 22.0  | 15.7  | 12.5 |
|                                                                                  | 2%          | WB     | 10.0  | 9.5   | 11.7  | 14.7  | 18.2  | 20.6  | 21.5  | 21.5  | 18.9  | 16.4  | 12.7  | 10.7 |
|                                                                                  |             | MCDB   | 11.0  | 10.8  | 15.1  | 20.4  | 23.3  | 26.4  | 28.2  | 27.8  | 23.8  | 19.0  | 14.0  | 11.7 |
|                                                                                  | 5%          | WB     | 8.6   | 8.6   | 10.6  | 13.2  | 16.7  | 19.4  | 20.3  | 20.2  | 17.6  | 15.2  | 11.6  | 9.5  |
|                                                                                  |             | MCDB   | 9.6   | 10.0  | 13.5  | 18.1  | 21.2  | 24.7  | 26.5  | 25.7  | 21.3  | 17.1  | 12.7  | 10.4 |
|                                                                                  | 10%         | WB     | 7.3   | 7.5   | 9.5   | 11.9  | 15.4  | 18.2  | 19.3  | 19.0  | 16.3  | 14.1  | 10.5  | 8.3  |
|                                                                                  |             | MCDB   | 8.3   | 8.9   | 11.8  | 15.7  | 19.6  | 23.0  | 24.6  | 23.4  | 19.3  | 15.6  | 11.5  | 9.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.5   | 7.3   | 9.9   | 12.1  | 11.7  | 11.8  | 11.7  | 12.0  | 11.4  | 9.3   | 6.1   | 5.4  |
|                                                                                  |             | MCDBR  | 6.1   | 9.3   | 15.3  | 17.5  | 17.0  | 16.7  | 16.3  | 17.1  | 15.9  | 11.7  | 7.3   | 5.5  |
|                                                                                  |             | MCWBR  | 5.3   | 7.0   | 10.0  | 10.4  | 9.8   | 9.1   | 7.9   | 9.0   | 9.7   | 8.1   | 5.8   | 4.8  |
|                                                                                  | 5% WB       | MCDBR  | 5.8   | 7.9   | 12.0  | 15.6  | 14.5  | 14.8  | 14.4  | 15.4  | 13.8  | 10.4  | 6.3   | 5.3  |
|                                                                                  |             | MCWBR  | 5.3   | 6.3   | 8.3   | 9.5   | 8.7   | 8.4   | 7.3   | 8.5   | 8.7   | 7.4   | 5.3   | 4.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.315  | 0.338 | 0.370 | 0.400 | 0.411 | 0.410 | 0.409 | 0.403 | 0.372 | 0.357 | 0.331 | 0.311 |      |
|                                                                                  | taud        | 2.425  | 2.381 | 2.296 | 2.232 | 2.238 | 2.267 | 2.266 | 2.309 | 2.376 | 2.414 | 2.423 | 2.397 |      |
|                                                                                  | Ebn at Noon | 731    | 800   | 830   | 842   | 848   | 850   | 844   | 831   | 821   | 765   | 700   | 674   |      |
|                                                                                  | Edh at Noon | 63     | 84    | 107   | 127   | 132   | 129   | 128   | 117   | 98    | 80    | 63    | 57    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.80   | 1.52  | 2.71  | 4.23  | 5.00  | 5.53  | 5.26  | 4.49  | 3.43  | 1.98  | 0.93  | 0.64  |      |
|                                                                                  | RadStd      | 0.13   | 0.28  | 0.38  | 0.52  | 0.55  | 0.51  | 0.55  | 0.50  | 0.35  | 0.17  | 0.11  | 0.11  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (66 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page